



Certificate

We hereby confirm that

Esperanza Buitrago Díaz

has completed the intensive program

Cyber Security Specialist

with 720 hours of programming practice (corresponds to 960 lessons) successfully.
The course has been delivered remotely.

Participation from 06.10.2025 until 05.02.2026

Franz Wilhelmy
Program Manager

Petra Kischkewitz
Managing Director

Dominic Singaraj
Lead Coach



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Curriculum Cyber Security

In 960 lessons the students have learned, discussed and practiced the following content in several tasks and projects:

Basics of IT Fundamentals

Hardware, devices, peripherals
Windows, Linux, macOS basics
Command-line tools & OS installation
IT troubleshooting & customer support

Networking Concepts

OSI & TCP/IP models
Ports, protocols (IPv4 and IPv6), services
Wireless networking standards
Switching concepts: VLANs, trunking, STP, port security.
Routing fundamentals: static vs dynamic routing
SOHO networking

Network Infrastructure

Routers, switches, firewalls, WAPs
Cabling types & connectors
Cloud networking & virtualization
VPNs: site-to-site, client-to-site, SSL vs IPsec tunnels.

Network Operations

Network documentation & diagrams
Monitoring tools (SNMP, Syslog, NetFlow)
Performance optimization

Network Security

Firewalls, IDS/IPS, and proxy servers.
Secure network design
Common attacks & mitigations
Access control & authentication
Wireless security standards: WPA2, WPA3, RADIUS, and EAP methods.

Network Troubleshooting

Troubleshooting methodology
Command-line utilities (ping, traceroute, nslookup)
Hardware & software diagnostics

Cybersecurity Fundamentals

CIA Triad (Confidentiality, Integrity, Availability).
Attack vectors: phishing, malware, ransomware
Cybersecurity frameworks: GDPR, ISO/IEC 27001, and CIS Controls.

Identity & Access Management (IAM)

Authentication, Authorization, and Accounting Framework
Authentication factors: passwords, biometrics, tokens
Authorization models: DAC, MAC, RBAC, ABAC
Single sign-on (SSO), Multi-Factor Authentication (MFA), least privilege, PAM

Cryptography

Symmetric and asymmetric encryption algorithms (AES, RSA, ECC)
Hashing methods (MD5, SHA) and digital signatures
Key management and PKI infrastructure
Secure communications: TLS, SSL, HTTPS, and VPN encryption

Risk Management & Compliance

Threat modeling, risk assessments, and vulnerability management
Security policies: Acceptable Use, BYOD, Data Classification
Regulatory requirements: GDPR, HIPAA, PCI-DSS
Business continuity and disaster recovery planning

Application & Host Security

Secure coding practices
Endpoint hardening and patch management
Secure cloud practices: SaaS, PaaS, IaaS models
Mobile device security

Security Operations & Monitoring

Security Operations Center (SOC) functions and SIEM tools
Log analysis & correlation

Threat Management & Vulnerability Management

Cyber threat intelligence (CTI)
Threat hunting & threat modeling.
Vulnerability scanning (OpenVAS)
Patch prioritization & remediation
Prioritizing vulnerabilities with CVSS scoring.

Incident Response

Preparation, detection, containment, eradication
Digital forensics basics
Malware analysis and reverse engineering basics

Compliance & Reporting

Compliance auditing for PCI-DSS, HIPAA, GDPR.
Incident reporting formats
Security documentation & communication

Red Team/Blue Team Simulation

Conduct penetration testing on a simulated enterprise
Use tools: Nmap, Metasploit, Hydra
Web application testing (OWASP Top 10)
Wireless & network penetration testing
VAPT-style reporting (scope, methodology, findings, CVSS, PoC, mitigations)